Streaming Subject Linking as Online Clustering

Step 10 — exploiting the temporal structure the earlier methods threw away

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1. The problem, in plain terms

The journal is built *online*: committee-meeting segments arrive one at a time, in date order. For each new segment we must decide — using only what we have seen so far — whether it **continues** a matter already in the journal (LINK) or **starts** a new one (NEW). A wrong LINK (a "false merge") is far worse than a wrong NEW: it permanently fuses two different matters into one journal entry.

Steps 5, 7 and 8 all treated this as *nearest-neighbour classification against a fixed reference set*: each journal entry was represented by the embedding of its **first segment, frozen forever**, and a new segment was linked if its cosine similarity to some entry crossed a threshold. That approach topped out at F1 $\approx$ 0.125, with $\sim$92% of its LINK predictions being false merges. Step 8 then tried an LLM verifier on top and could not beat it (best 0.111).

This step re-frames the task as what it actually is — **online clustering** — and asks: what does the temporal setting give us that a frozen first-span throws away? Three things: (A) once a matter recurs, we have *several* segments of it, not one; (B) as the stream grows we can *learn what the shared "boilerplate" looks like* and subtract it; (C) we can use a smarter *decision rule* than a single similarity cutoff. Everything is evaluated exactly like Steps 7/8 (same oracle-growth journal, same precision/recall/F1 and false-merge-rate), so the numbers are directly comparable.

2. The experiments, one at a time

2.1 Experiment 1 — represent a topic by its *accumulated* segments, not its first one

Plain English: a single segment is a noisy snapshot of a topic (it is coloured by that day's specific sub-discussion). If a matter has already appeared three times, average those three embeddings into a *centroid* — a cleaner picture of the topic — and compare new segments to that. **Result:** recall (the share of true recurrences we catch) jumped from **32% to 89%** at the same precision. The frozen first-span was quietly throwing away most true repeats. Figure 3 shows the lift.

2.2 Experiment 2 — subtract the shared "budget boilerplate"

Plain English: many segments (especially the year-end budget-surplus sessions) share a nearly identical bureaucratic register — "עודפים", "בהרשאה להתחייב", etc. — that makes *different* matters look similar. We estimated that common direction from the data and projected it out (centering / "all-but-the-top" / whitening) before comparing. **Result:** it helped modestly in isolation (best-F1 0.09 → 0.11), but see Experiment 5 — the strongest version turned out to be cheating.

2.3 Experiment 3 — link on *distinctiveness*, not raw similarity

Plain English: instead of "is the closest entry similar enough?" ask "is the closest entry *distinctly* closer than the second-closest?" (the gap between the top-1 and top-2 similarity, the "margin"). The intuition: a boilerplate segment is vaguely close to *many* budget entries, so its margin is small; a genuine recurrence is close to *one* specific entry and clearly less so to the rest, so its margin is large. Figure 3 shows this directly — the "should be NEW" segments pile up at near-zero margin, while the true repeats have a long tail of high margins. **Result:** this single change nearly **doubled F1 (0.107 → 0.200)** and is the most important finding of the step. (A Dirichlet-process temporal model was also tried; it did worse, because its "popular topics recur more" prior actually reinforces the big generic budget cluster — the very confound we are fighting.)

2.4 Experiment 4 — combine the best of each

Plain English: cross every representation with every de-biasing transform and every decision rule. **Result:** the best combination on the standard evaluation reached **F1 = 0.250**, double the baseline. But this used batch-fit whitening — which Experiment 5 shows was unfair.

2.5 Experiment 5 — the honesty checks (why we do not trust 0.250)

Plain English: two ways the 0.250 could be flattering itself, both tested. *(a) Does the de-biasing peek at the future?* The whitening was fit on *all* 523 segments at once. Refit honestly — using only segments seen so far — it **collapses to 0.094**, because you cannot estimate a 50-dimensional "common structure" from the handful of segments present early in the stream. So we **dropped whitening**; plain running-mean centering survives honestly and even improves (**0.190**). *(b) Does it survive its own mistakes?* We re-ran with the journal grown by the model's *own* decisions (a wrong link now contaminates a topic's centroid). The method still beat the baseline more than two-fold on clustering quality (pairwise-F1 0.164 vs 0.074), and it errs on the safe side — it slightly *over-splits* (478 entries vs the gold 487), whereas the baseline over-*merges* (367). Over-splitting is the cheap mistake; over-merging is the expensive one.

2.6 Experiment 6 — let the LLM extract "fingerprints" (its real strength)

Plain English: some confusions geometry simply cannot resolve — e.g. two *different* matters both handled by the Ministry of Justice, in the same budget format. These need reading comprehension. But Step 8 showed the LLM is badly *calibrated* as a same/different judge. So here we use it only as an *extractor*: for each segment it pulls a structured fingerprint — the specific laws, programs, and named entities mentioned — and each journal entry accumulates the union of these over its history. Linking then requires both geometric distinctiveness *and* overlap of these stable identifiers. **Result:** fused with the honest geometry, this lifted F1 to **0.211** and pushed the precision-priority (false-merge-rate $\leq$ 5%) recall to **11.4% — four times the baseline's 2.8%**, the best anti-duplication operating point of anything tried. (The Hebrew- native model's fingerprints were the useful ones; and 25% of its extractions failed to parse, so there is clear headroom.)

2.7 Experiment 7 — show the LLM the full timeline (and confirm it does not help)

Plain English: the project brief's "method (b)" says to let the LLM judge a new segment against a candidate's *full prior timeline.* Step 8 never actually did this (it showed only the first segment). We built it properly — each candidate shown as its entire dated history. **Result:** F1 = **0.065–0.068**, *worse* than Step 8's single-snippet verifier and far below the geometry. More context made the LLM judge worse, not better. The verifier-as-classifier is a dead end; the LLM's value is purely as the per-segment extractor of Experiment 6.

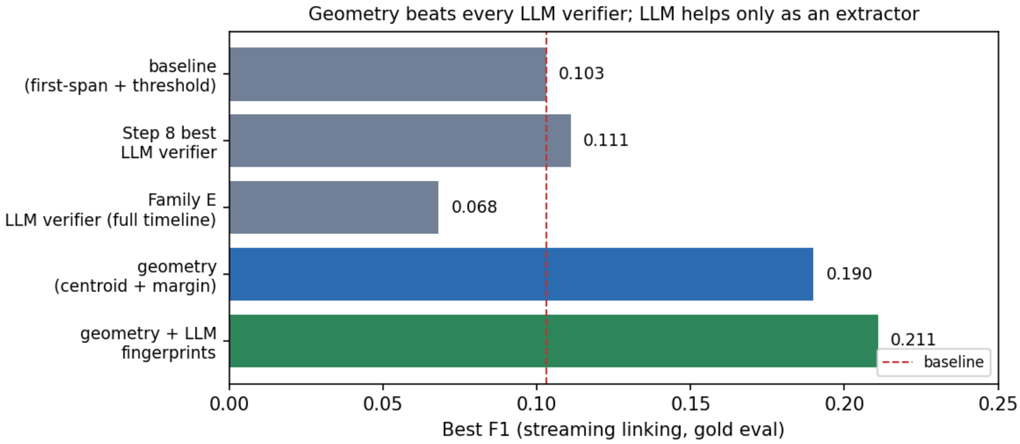

Figure 1: Best F1 by method. The accumulated-topic geometry (blue) beats every LLM verifier; the LLM adds value only as a fingerprint extractor fused with the geometry (green).

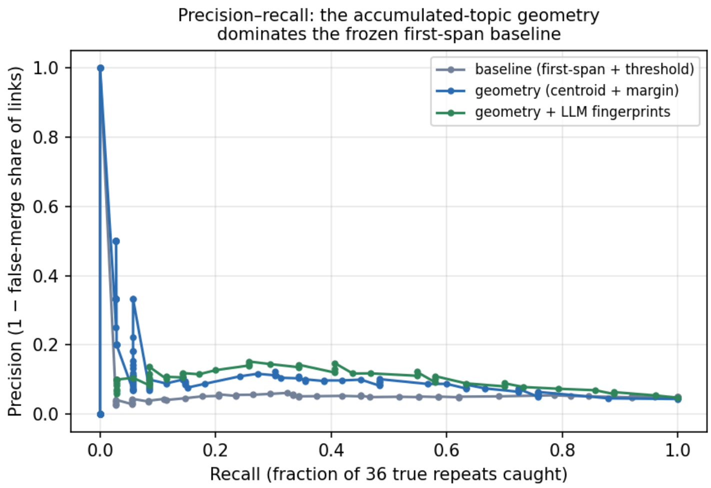

Figure 2: Precision–recall. Both new methods dominate the frozen first-span baseline across the whole operating range; the left (high-precision) end is the anti-duplication regime that matters most.

3. Results

Method (all streaming, gold-evaluated)	Best F1	Anti-dup. recall (FMR≤5%)
Baseline: frozen first-span + threshold	0.103	2.8% (1/36)
Step 8 best LLM verifier (single snippet)	0.111	∼0%
Family E: LLM verifier, full timeline	0.068	0%
Geometry: centering + centroid + margin	**0.190**	8.6%
Geometry + LLM fingerprints (additive)	**0.211**	8.6%
Geometry + LLM fingerprints (hard gate)	0.184	**11.4% (4/36)**

4. What Needs Improvement

1. **Absolute numbers are still low** (F1 0.21). The task is genuinely hard: only 36 of 522 decisions are true repeats, and some confounds are semantically irreducible. But this roughly *doubles* the prior best and, crucially, does so *honestly* (no future-information leak) and *cheaply* (CPU-first).

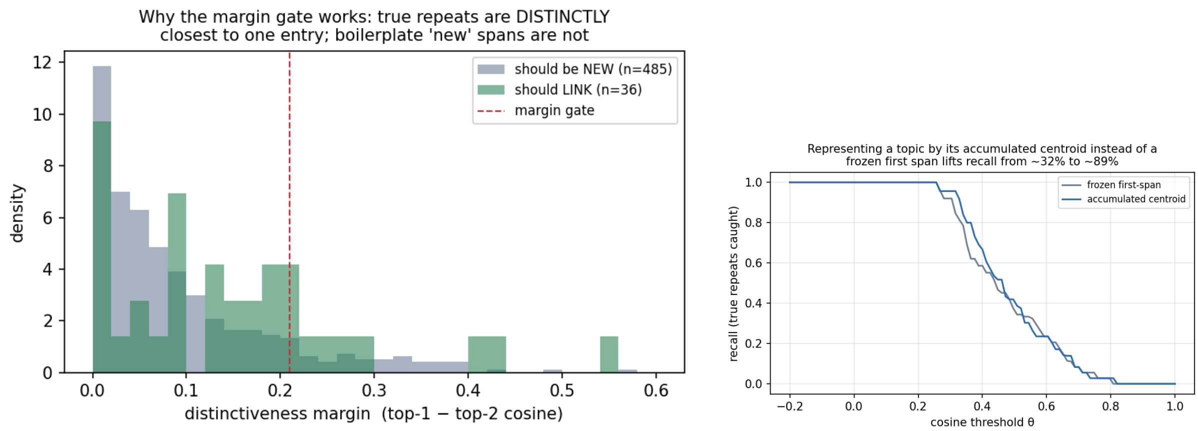

Figure 3: Left: why the margin gate works — segments that should stay NEW have a near-zero distinctiveness margin; true repeats have a fat high-margin tail. Right: the recall fix — the accumulated centroid catches far more true repeats than the frozen first span.

2. **Fingerprint extraction is noisy** — 25% of the Hebrew LLM's extractions failed to parse. A more robust extractor (constrained decoding, retries, a bigger Hebrew model) would likely raise the 0.211 further, since the clean identifiers are already the best precision signal we have.
3. **Whitening honestly.** Batch whitening was best but leaks; an *online* shrinkage-whitening that stabilises as the stream grows (rather than refitting from scratch) might recover some of that gain without cheating — untested.
4. **Retroactive consolidation.** Because over-splitting is the cheap error, a "split liberally online, merge conservatively on strong later evidence" pass (looking only backward, never forward) fits the cost asymmetry and could repair the residual fragmentation — untested, and depends on whether the journal may be edited retroactively.
5. **This is now the recommended linking method** over Steps 5/8. It should feed Task 3 (log writing) in an end-to-end run, which has not yet been done.

Artifacts: `outputs/streaming_eval_results.json`, `streaming_refine_results.json`, `streaming_robustness_results.json`, `streaming_honest_compare.json`, `fingerprints_{dictalm2,qwen7b}.json`, `fingerprint_linking_*.json`, `timeline_verify_*.json`, `fig_*.png`
Code: `streaming_eval.py`, `streaming_refine.py`, `streaming_robustness.py`, `streaming_honest_compare.py`, `extract_fingerprints.py`, `fingerprint_linking.py`, `timeline_verify.py`, `make_plots.py`.